Software Engineering Lab 8

Selenium Testing – GearVault Equipment Rental System

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Date: 22 October 2026

Task: Provide any 5 test cases for the project. Execute any two test cases using Selenium, record the execution and upload the recording.

1. Introduction

GearVault is a web-based professional equipment rental system. Customers browse a catalog of cameras, audio, lighting and computer gear, book equipment for a date range, and track their reservations. Staff and managers have separate dashboards for dispatch, analytics, disputes and inventory. The application is deployed on AWS (CloudFront, S3, ECS and RDS PostgreSQL).

This lab tests the core user-facing flows with Selenium IDE: account sign-up, login (valid and invalid), catalog browsing and search, and role-based access control. Five test cases were written; all five were executed, and three of them (TC01, TC03, TC04) were recorded on video.

2. Test Environment

Item	Details
Application under test	https://d2wvsbj3dzkosr.cloudfront.net
Deployment	AWS – CloudFront + S3 (frontend), ECS (Flask API), RDS PostgreSQL
Testing tool	Selenium IDE 4.0.1 (desktop app)
Browser	Chromium (built into Selenium IDE)
Operating system	Windows 11
Test project file	GearVault.side (5 test cases, one suite)
Locator strategy	CSS selectors and XPath

Each test starts as a logged-out guest by clearing the browser session. Tests that need an account create one with a unique email (selenium.<timestamp>@test.com), so every test is independent and can be re-run any number of times. The recorded tests include a 1-second pause after each step so the actions are easy to follow on video.

3. Test Cases

ID	Title	Type	Recorded
TC01	New customer sign up	Positive	Yes
TC02	Login with valid credentials	Positive	No
TC03	Login with wrong password is rejected	Negative	Yes
TC04	Guest browses and searches the catalog	Positive	Yes
TC05	Role-based access control	Negative / Security	No

3.1 TC01 – New customer sign up

Test Case ID	TC01
Title	New customer sign up
Type	Positive / Functional
Objective	A new visitor can create an account and is logged in as a customer.
Precondition	User is not logged in.
Test Data	Full name: Selenium Tester; Email: selenium.<timestamp>@test.com (unique each run); Password: Test@1234

Step	Action	Expected Result
1	Open /login	"Sign In to GearVault" page is shown
2	Click "Create account"	"Create Your GearVault Account" page is shown
3	Enter full name, unique email and password	Fields accept the input
4	Click "Create Account"	User is redirected to the Equipment Inventory dashboard
5	Check the navigation bar	Shows the new email, a CUSTOMER role badge and a "Reservations" link

Actual Result	Account created; dashboard shows the new email and the CUSTOMER badge.
Status	**Pass (recorded)**

3.2 TC02 – Login with valid credentials

Test Case ID	TC02
Title	Login with valid credentials
Type	Positive / Functional
Objective	A registered user can sign in with the correct email and password.
Precondition	A registered account exists (the test registers one, then signs out).
Test Data	Email: selenium.<timestamp>@test.com; Password: Test@1234

Step	Action	Expected Result
1	Register a new account, then click "Sign Out"	Sign In page is shown
2	Enter the same email and password Test@1234	Fields accept the input
3	Click "Sign In"	User is redirected to "Equipment Inventory"
4	Check the navigation bar	Shows the user's email and the CUSTOMER badge

Actual Result	User signed in and was redirected to the dashboard with the correct email and role.
Status	**Pass**

3.3 TC03 – Login with wrong password is rejected

Test Case ID	TC03
Title	Login with wrong password is rejected
Type	Negative / Security
Objective	Wrong credentials are rejected and no session is created.
Precondition	A registered account exists (the test registers one, then signs out).
Test Data	Email: selenium.<timestamp>@test.com; Password: WrongPass999

Step	Action	Expected Result
1	Register a new account, then click "Sign Out"	Sign In page is shown
2	Enter the correct email with password WrongPass999	Fields accept the input
3	Click "Sign In"	Red error message "Invalid email or password" is shown
4	Check the navigation bar	No user badge: the user is not logged in

Actual Result	"Invalid email or password" was displayed and the user stayed logged out.
Status	**Pass (recorded)**

3.4 TC04 – Guest browses and searches the catalog

Test Case ID	TC04
Title	Guest browses and searches the catalog
Type	Positive / Functional
Objective	Anyone can browse, filter and search the equipment catalog without logging in.
Precondition	User is not logged in. Catalog contains "Canon EOS R6" and "Sony A7 IV" under Cameras.
Test Data	Category: Cameras; Search text: Sony A7

Step	Action	Expected Result
1	Click "Inventory"	Equipment Inventory page with item cards
2	Select Category "Cameras"	"Canon EOS R6" is listed
3	Type "Sony A7" in Search Equipment	"Sony A7 IV" is listed and "Canon EOS R6" disappears
4	Click "Sony A7 IV"	Item detail page with the heading "Sony A7 IV" opens

Actual Result	Filter and search narrowed the list correctly; the Sony A7 IV detail page opened.
Status	**Pass (recorded)**

3.5 TC05 – Role-based access control

Test Case ID	TC05
Title	Role-based access control
Type	Negative / Security
Objective	Protected pages cannot be reached by guests or by users with the wrong role.
Precondition	User is not logged in.
Test Data	New customer account (selenium.<timestamp>@test.com / Test@1234)

Step	Action	Expected Result
1	As a guest, open /bookings	User is redirected to the Sign In page
2	Register a new customer account	Logged in as CUSTOMER
3	Check the navigation bar	No "Analytics" link (manager only)
4	Open /analytics directly	"403 — Access Denied" page is shown

Actual Result	Guest was redirected to login; customer saw no manager menu and got "403 — Access Denied".
Status	**Pass**

4. Execution Summary

ID	Test Case	Execution	Result
TC01	New customer sign up	Selenium IDE – recorded	Pass
TC02	Login with valid credentials	Selenium IDE	Pass
TC03	Login with wrong password is rejected	Selenium IDE – recorded	Pass
TC04	Guest browses and searches the catalog	Selenium IDE – recorded	Pass
TC05	Role-based access control	Selenium IDE	Pass

Total: 5 test cases | 5 passed | 0 failed | 3 recorded (2 were required)

5. Execution Recording

The three recorded executions are combined into one video. Each test is introduced by a 3-second title card showing the test case number, its description and the expected result.

Video file	GearVault_Selenium_Tests.mp4 (1 min 14 s)
Video link	[Paste the uploaded video link here]
Timestamps	0:00 – TC01 New customer sign up | 0:25 – TC03 Wrong password | 0:57 – TC04 Catalog search

6. Screenshots

Each screenshot shows the Selenium IDE window: the 5 test cases and the command list (completed steps in green) on the left, and the GearVault application being driven on the right.

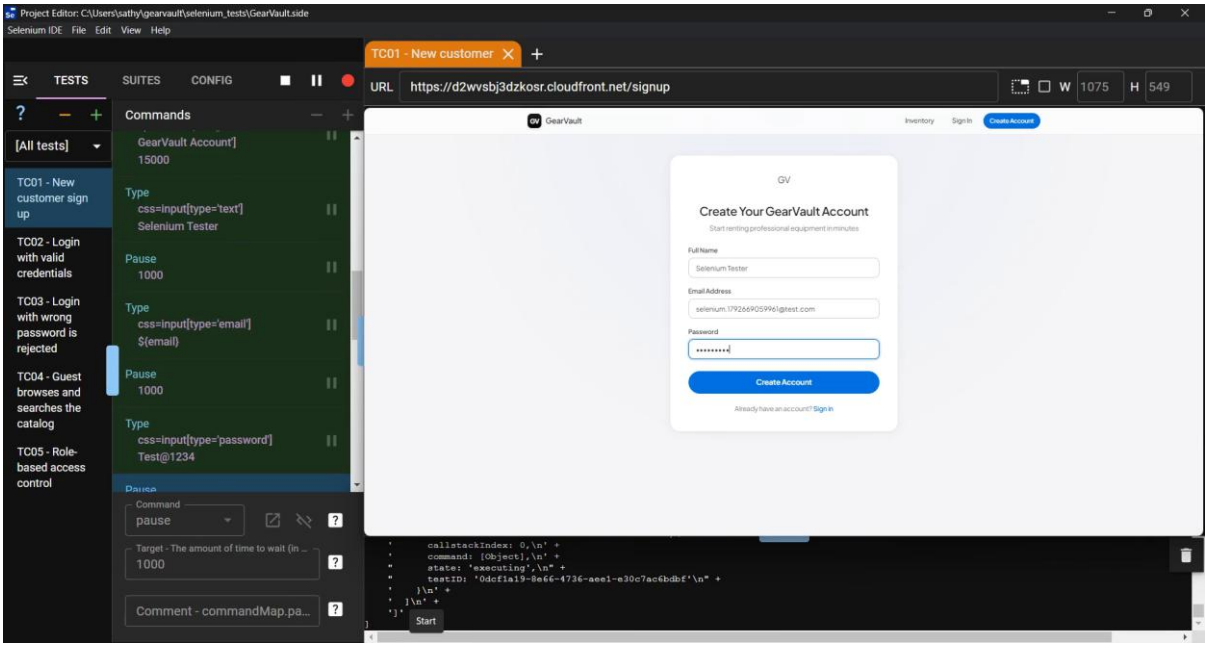


Figure 1 – TC01 in progress: Selenium IDE filling in the sign-up form step by step

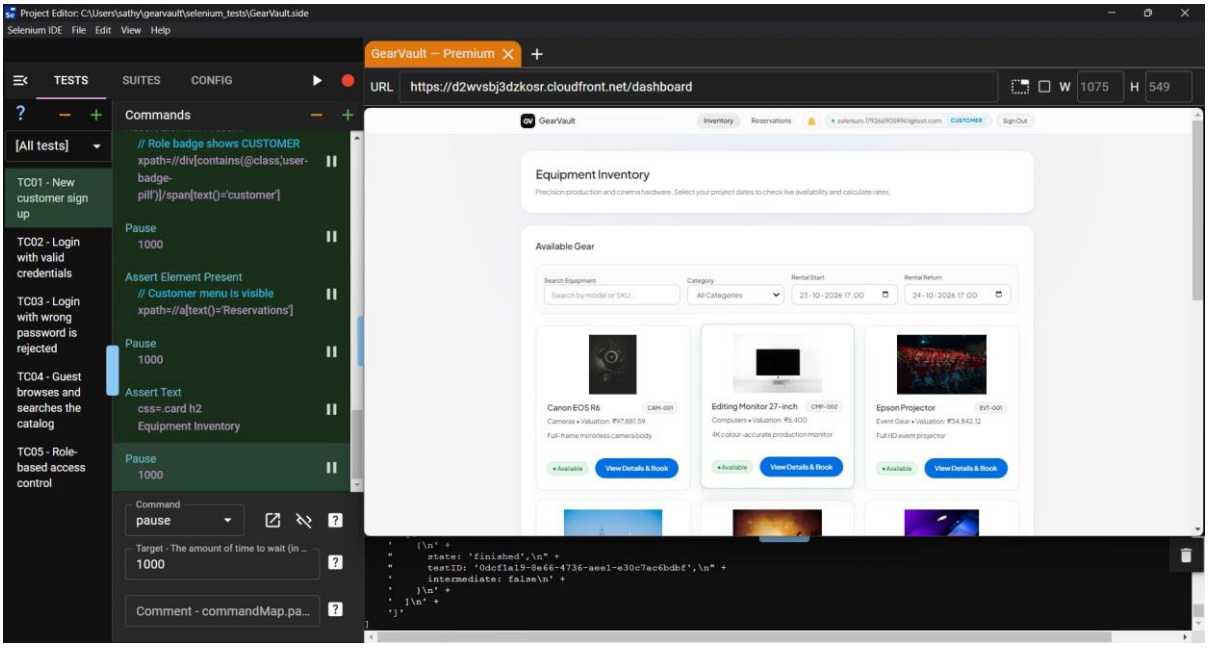


Figure 2 – TC01 passed: new customer logged in, navbar shows the email and the CUSTOMER badge

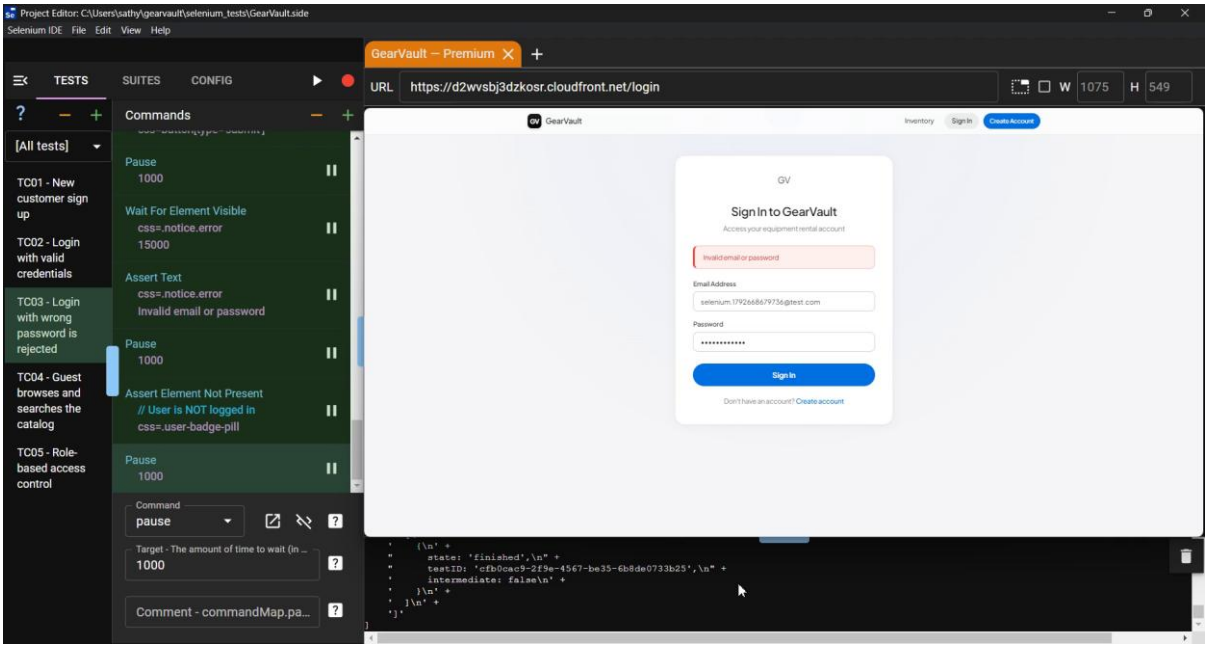


Figure 3 – TC03 passed: wrong password rejected with "Invalid email or password"

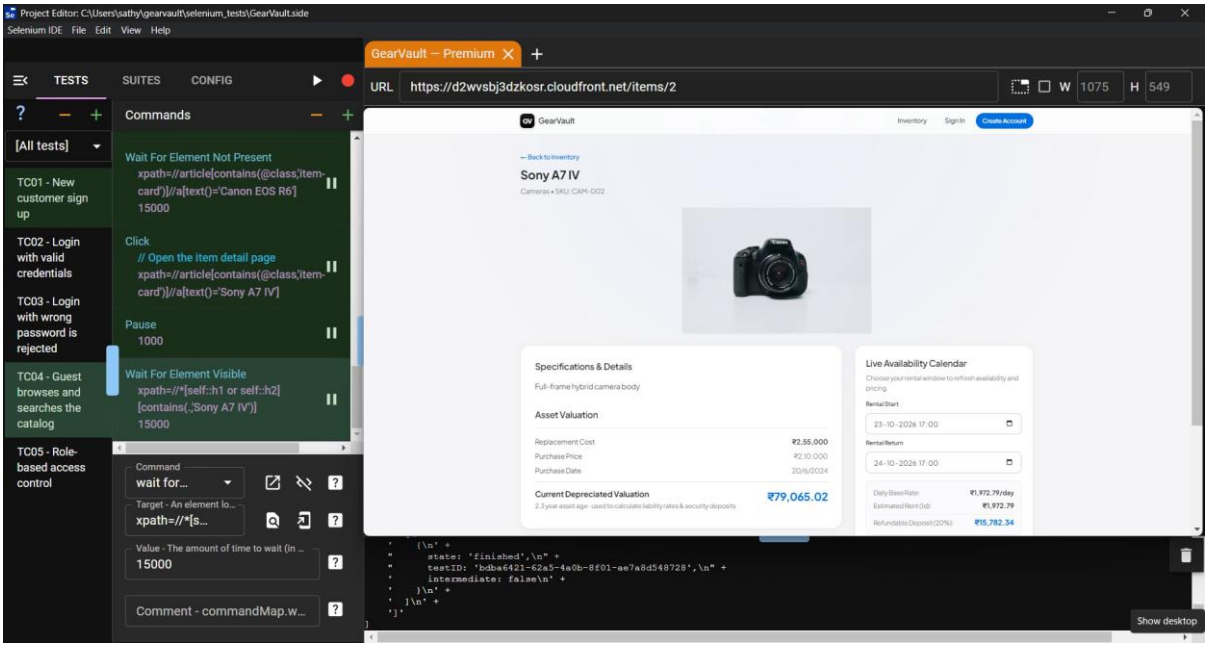


Figure 4 – TC04 passed: Cameras filter + "Sony A7" search opened the Sony A7 IV detail page

7. Issues Found and Fixed

- **linkText locator failure:** In Selenium IDE 4, steps using the linkText= locator (for example clicking "Create account") failed intermittently with "Cannot read properties of undefined (reading 'findElements')". All linkText locators were replaced with equivalent XPath locators (//a[text()='Create account']), after which every test passed consistently.
- **Asynchronous loading:** The category dropdown loads after the page, so selecting "Cameras" immediately caused a retry. A waitForElementPresent step was added before the selection.
- **setSpeed not supported:** Selenium IDE 4 does not support the setSpeed command, so pause steps were used to slow down the recorded runs.

8. Conclusion

Five test cases covering sign-up, valid and invalid login, catalog search and role-based access control were designed and automated with Selenium IDE against the live AWS deployment of GearVault. All five passed. Three executions were recorded and combined into a single video. The tests confirmed that authentication works correctly, invalid credentials are rejected, the public catalog filter and search work for guests, and protected pages are restricted by role.